SCALARIZATION METHODS WITH A CHARACTERIZING FUNCTION OF AN ORDERING CONE IN SET OPTIMIZATION

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1. Introduction

In convex analysis and optimization theory, Fan-Takahashi minimax inequality plays a key role to solve equilibrium problems and other problems. Let X be a nonempty compact convex subset of a Hausdorff topological vector space and $f\colon X \times X \to \mathbb{R}$. Fan-Takahashi minimax inequality is: if f satisfies the following conditions:

(a) for each fixed $y \in X$, $f(\cdot, y)$ is lower semicontinuous,
(b) for each fixed $x \in X$, $f(x, \cdot)$ is quasi concave,
(c) $f(x, x) \leq 0$ for all $x \in X$,

then there exists $\bar{x} \in X$ such that $f(\bar{x}, y) \leq 0$ for all $y \in X$. In addition, a number of researchers, including Georgiev and Tanaka, study Fan-Takahashi minimax inequalities for set-valued maps. In this talk, we explain set-valued Fan-Takahashi minimax inequalities generalized by scalarization functions, and consider relations between scalarization functions for set-valued Fan-Takahashi minimax inequalities and characterizing functions of the ordering cone C [1] defined by

$$f(y) \leq 0 \Leftrightarrow y \in -C \quad \text{for all } y \in Y,$$

where C is a convex cone in a real topological vector space Y and $f : Y \to \mathbb{R} \cup \{\pm\infty\}$.

References

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